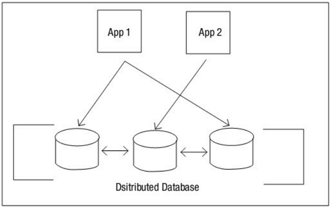


Figure 1: Distributed data storage

The greatest deployment of traditional RDBMS is primarily in enterprises, for everything the enterprise does, whether storing customer information, internal financials, employee information or anything else. Is such data growing? Yes, so big-data issues are relevant to them. So why not NoSQL?

Even though there are varieties of NoSQL offerings, they are typically characterised by lack of SQL support, and non-adherence to ACID (Atomicity, Consistency, Isolation and Durability) properties. (Here, we will not go into the technical reasons for this.) So while NoSQL could help enterprises manage large distributed data, enterprises cannot afford to lose the ACID properties—which were the key reasons for choosing an RDBMS in the first place. Also, since NoSQL solutions don't provide SQL support, which most current enterprise applications require, this pushes enterprises away from NoSQL. Hence, a new set of data-management solutions are emerging to address large data OLTP concerns, without sacrificing ACID and SQL interfaces.

## Why is traditional OLTP insufficient?

We have already discussed the fact that there is an explosion in the OLTP data space. For example, more widely used social network sites (like Facebook and Linkedin) lead to larger OLTP requirements. Each user of such sites requires credentials and user profile information to be stored, generally in some OLTP database, even though the actual user data is stored in a NoSQL data-store. Facebook's 750 million users require a very large OLTP database.

While growing OLTP data is a direct contributor, business requirements also force non-OLTP data to be managed as OLTP. Let's look at the example of how analytics (which is the quintessential OLAP application) has led to the scalability requirements of OLTP.

## Case study: Real-time analytics

In the earlier days, analytics were performed on historic data with specialised data warehouse solutions, mostly using an ETL (Extract, Transform and Load) approach. Data extracted from OLTP systems was fed to data warehouse systems capable of handling voluminous data. Real-time analytics is an approach that enables business users to get up-to-the-minute data by directly accessing business operational systems or feeding real-time transactions to analytics systems. For example, previously Google Analytics analysed past performance—but recently, Google launched Google Analytics: Real Time, which shows a set of new reports on what's happening on their site, as it happens.

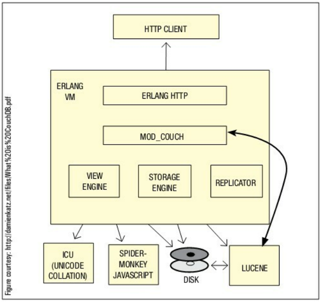


Figure 2: CouchDB architecture

Even though traditional OLTP systems provide ACID, they are not well equipped to handle the volume of data seen because of new innovative business scenarios like real-time analytics. A combination of traditional OLTP systems and analytics systems might under-utilise the analytics systems, due to scalability limitations and the performance of traditional OLTP systems.

## NewSQL

To address big-data OLTP business scenarios that neither traditional OLTP systems nor NoSQL systems address, alternative database systems have evolved, collectively named NewSQL systems. This term was coined by the 451 Group, in their now famous report, 'NoSQL, NewSQL and Beyond'. The term NewSQL was used to categorise these new alternative database systems. 451 Group's senior analyst, Matthew Aslett, clarified the meaning of the term NewSQL (in his blog) as follows: "NewSQL is our shorthand for the various new scalable/high-performance SQL database vendors. We have previously referred to these products as 'ScalableSQL' to differentiate them from the incumbent relational database products. Since this implies horizontal scalability, which is not necessarily a feature of all products, we adopted the term NewSQL in the new report. And to clarify, like NoSQL, NewSQL is not to be taken too literally: the new thing about the NewSQL vendors is the vendor, not the SQL. NewSQL is a set of various new scalable/high-performance SQL database vendors (or databases). These vendors have designed solutions